

Artificial intelligence

Artificial intelligence (AI) is the capability of computational systems to perform tasks typically associated with human intelligence, such as learning, reasoning, problem-solving, perception, and decision-making. It is a field of research in engineering, mathematics, and computer science that develops and studies methods and software that enable machines to perceive their environment and use learning and intelligence to take actions that maximise their chances of achieving defined goals. High-profile applications of AI include advanced web search engines, chatbots, virtual assistants, autonomous vehicles, play and analysis in strategy games (e.g., chess and Go), and content generation (e.g., text, images, audio, and videos). The traditional goals of AI research include learning, reasoning, knowledge representation, planning, natural language processing, and perception, as well as support for robotics. To reach these goals, AI researchers use techniques including state space search and mathematical optimisation, formal logic, artificial neural networks, and methods based on statistics, operations research, and economics. AI also draws upon psychology, linguistics, philosophy, neuroscience, and other fields. Some companies, such as OpenAI, Google DeepMind, and Meta, aim to create artificial general intelligence (AGI)—AI that can complete nearly any cognitive task at least as well as a human. Artificial intelligence was founded as an academic discipline in 1956. The field went through multiple cycles of optimism throughout its history, followed by periods of disappointment and loss of funding, known as AI winters. Funding and interest increased substantially after 2012, when graphics processing units (GPUs) started being used to accelerate neural networks, and deep learning outperformed previous AI techniques. This growth accelerated further after 2017 with the transformer architecture. In the 2020s, an AI boom coincided with advances in generative AI, which became widespread and allowed for the creation and modification of media. In addition to AI safety and unintended consequences and harms from the use of AI, ethical concerns, AI's long-term effects, environmental effects, and potential existential risks have prompted discussions of AI regulation.

Goals The general problem of simulating (or creating) intelligence has been broken down into subproblems. These consist of specific traits or capabilities that researchers expect an intelligent system to display. The traits described below have received the most attention and cover the scope of AI research.

Reasoning and problem-solving Early researchers developed algorithms that imitated step-by-step reasoning that humans use when solving puzzles or making logical deductions. By the late 1980s and 1990s, methods were developed for dealing with uncertain or incomplete information, employing concepts from probability and economics. Many of these algorithms were insufficient for solving large reasoning problems because they experienced a "combinatorial explosion", meaning they become exponentially slower as the problems grow. Even humans rarely use the step-by-step deduction that early AI research could model. Humans solve most of their problems using fast, intuitive judgments. Reasoning models, a type of large language model (LLM) trained to generate intermediate chains of thought, emerged in 2024 and allowed improved performance on complex problems in mathematics and coding. These models can produce incorrect outputs or "hallucinations," unlike symbolic reasoning systems.

Knowledge representation

AI programs use knowledge to answer questions intelligently and make deductions about real-world facts. Formal knowledge representation and knowledge engineering use symbols to represent words, concepts and things in the world. A knowledge base is a body of knowledge represented in a form that can be used by a program. An ontology is the set of objects, relations, concepts, and properties used by a particular domain of knowledge. Formal knowledge has been studied extensively since the 1970s and researchers have developed formalisms for a wide variety of difficult domains. The symbolic approach has difficulty

with several problems: the breadth of commonsense knowledge (the set of atomic facts the average person knows is enormous), the sub-symbolic form of most commonsense knowledge (much of what people know is not represented as "facts" or "statements" they can express verbally), and knowledge acquisition (the problem of obtaining knowledge for AI applications). Large language models (and some other AI programs developed since 2012) do not require explicit, symbolic knowledge. They acquire knowledge by being trained on the combined text of millions of books and billions of websites. Modern AI can also learn about a domain by running experiments (as when AlphaZero learns game strategy by playing against itself). Machine learning solves the problems of general knowledge, commonsense knowledge and knowledge acquisition, however, this approach has struggled with accurate recall and valid reasoning.

Planning and decision-making An "agent" is any entity (artificial or not) that perceives and takes actions in the world. A rational agent has goals or preferences and takes actions to make them happen. In automated planning, the agent has a specific goal. In automated decision-making, the agent has preferences—there are some situations it would prefer to be in, and some situations it is trying to avoid. The decision-making agent assigns a number to each situation (called its "utility") that measures how much the agent prefers it. For each possible action, it can calculate the "expected utility": the utility of all possible outcomes of the action, weighted by the probability that the outcome will occur. It can then choose the action with the maximum expected utility. In classical planning, the agent knows exactly what the effect of any action will be. In most real-world problems, however, the agent may not understand its current situation with certainty (it is "unknown" or "unobservable") and it may not know for certain what will happen after each possible action (it is not "deterministic"). It must choose an action by making a probabilistic guess and then reassess the situation to see if the action had the desired effect. Alongside thorough testing and improvement based on previous decisions, having an explanation for why the agent took certain decisions is a way to build trust, especially when the decisions have to be relied upon. In some problems, the agent's preferences may be uncertain, especially if there are other agents or humans involved. These preferences may be learned (e.g., with inverse reinforcement learning), or the agent can seek information to improve them. Information value theory can be used to weigh the value of exploratory or experimental actions. The space of possible future actions and situations is typically intractably large, so the agents must take actions and evaluate situations while being uncertain of the outcome. A Markov decision process has a transition model that describes the probability that a particular action will change the state in a particular way and a reward function that supplies the utility of each state and the cost of each action. A policy associates a decision with each possible state. The policy could be calculated (e.g., by policy iteration), determined by a heuristic, or learned. Game theory describes the rational behaviour of multiple interacting agents and is used in AI programs that make decisions involving other agents.

Learning Machine learning is the study of programs that can improve their performance on a given task automatically. It has been a part of AI from the beginning.

There are several kinds of machine learning:

Unsupervised learning analyses a stream of data, finds patterns, and makes predictions without any other guidance. Supervised learning requires labelling the training data with the expected answers, and comes in two main varieties: classification (where the program must learn to predict what category the input belongs in) and regression (where the program must deduce a numeric function based on numeric input). Reinforcement learning is when the agent is rewarded for good responses and punished for bad ones. The agent learns to choose responses that are classified as "good". Transfer learning is when the knowledge gained from one problem is applied to a new problem. Deep learning is a type of machine learning that

runs inputs through biologically inspired artificial neural networks for all of these types of learning. Computational learning theory can assess learners by computational complexity, sample complexity (how much data is required), or other notions of optimisation.

Natural language processing Natural language processing (NLP) allows programs to read, write, and communicate in human languages. Specific problems include speech recognition, speech synthesis, machine translation, information extraction, information retrieval, and question answering. Early work, based on Noam Chomsky's generative grammar and semantic networks, had difficulty with word-sense disambiguation unless restricted to small domains called "micro-worlds" (due to the common sense knowledge problem). British linguist and philosopher Margaret Masterman believed it was meaning and not grammar that was the key to understanding languages, and that dictionaries and especially thesauri should be the basis of computational language structure. Modern deep learning techniques for NLP include word embedding (representing words, typically as vectors encoding their meaning), transformers (a deep learning architecture using an attention mechanism), and others. In 2019, generative pre-trained transformer (or "GPT") language models began to generate coherent text. By 2023, these models were able to get human-level scores on the bar exam, SAT (Scholastic Assessment Test), GRE (Graduate Record Examination), and many other real-world applications.

Perception Machine perception is the ability to use input from sensors (such as cameras, microphones, wireless signals, active lidar, sonar, radar, and tactile sensors) to deduce aspects of the world. Computer vision is the ability to analyse visual input. The field includes speech recognition, image classification, facial recognition, object recognition, object tracking, and robotic perception.

Social intelligence

Affective computing is a field that comprises systems that recognise, interpret, process, or simulate human affect (feeling, emotion, and mood). For example, some virtual assistants are programmed to speak conversationally or even banter humorously; it makes them appear more sensitive to the emotional dynamics of human interaction, or to otherwise facilitate human-computer interaction. However, this tends to give naïve users an unrealistic conception of the intelligence of existing computer agents. Moderate successes related to affective computing include textual sentiment analysis and, more recently, multimodal sentiment analysis, wherein AI classifies the effects displayed by a videotaped subject.

General intelligence A machine with artificial general intelligence (AGI) would be able to solve a wide variety of problems with breadth and versatility similar to human intelligence.

Techniques AI research uses a wide variety of techniques to accomplish the goals above.

Search and optimisation There are two different kinds of search used in AI: state space search and local search:

State space search State space search searches through a tree of possible states to try to find a goal state. For example, planning algorithms search through trees of goals and subgoals, attempting to find a path to a target goal, a process called means-ends analysis. Simple exhaustive searches are rarely sufficient for most real-world problems: the search space (the number of places to search) quickly grows to astronomical numbers. The result is a search that is too slow or never completes. "Heuristics" or "rules of thumb" can help prioritise choices that are more likely to reach a goal. Adversarial search is used for game-playing programs, such as chess or Go. It searches through a tree of possible moves and countermoves, looking for a winning position.

Local search

Local search uses mathematical optimisation to find a solution to a problem. It begins with some form of guess and refines it incrementally. Gradient descent is a type of local search

that optimises a set of numerical parameters by incrementally adjusting them to minimise a loss function. Variants of gradient descent are commonly used to train neural networks, through the backpropagation algorithm. Another type of local search is evolutionary computation, which aims to iteratively improve a set of candidate solutions by "mutating" and "recombining" them, selecting only the fittest to survive each generation. Distributed search processes can coordinate via swarm intelligence algorithms. Two popular swarm algorithms used in search are particle swarm optimisation (inspired by bird flocking) and ant colony optimisation (inspired by ant trails).

Logic Formal logic is used for reasoning and knowledge representation. Formal logic comes in two main forms: propositional logic (which operates on statements that are true or false and uses logical connectives such as "and", "or", "not" and "implies") and predicate logic (which also operates on objects, predicates and relations and uses quantifiers such as "Every X is a Y" and "There are some Xs that are Ys"). Deductive reasoning in logic is the process of proving a new statement (conclusion) from other statements that are given and assumed to be true (the premises). Proofs can be structured as proof trees, in which nodes are labelled by sentences, and children nodes are connected to parent nodes by inference rules. Given a problem and a set of premises, problem-solving reduces to searching for a proof tree whose root node is labelled by a solution of the problem and whose leaf nodes are labelled by premises or axioms. In the case of Horn clauses, problem-solving search can be performed by reasoning forwards from the premises or backwards from the problem. In the more general case of the clausal form of first-order logic, resolution is a single, axiom-free rule of inference, in which a problem is solved by proving a contradiction from premises that include the negation of the problem to be solved. Inference in both Horn clause logic and first-order logic is undecidable, and therefore intractable. However, backward reasoning with Horn clauses, which underpins computation in the logic programming language Prolog, is Turing complete. Moreover, its efficiency is competitive with computation in other symbolic programming languages. Fuzzy logic assigns a "degree of truth" between 0 and 1. It can therefore handle propositions that are vague and partially true. Non-monotonic logics, including logic programming with negation as failure, are designed to handle default reasoning. Other specialised versions of logic have been developed to describe many complex domains.

Probabilistic methods for uncertain reasoning

Many problems in AI (including reasoning, planning, learning, perception, and robotics) require the agent to operate with incomplete or uncertain information. AI researchers have devised a number of tools to solve these problems using methods from probability theory and economics. Precise mathematical tools have been developed that analyse how an agent can make choices and plan, using decision theory, decision analysis, and information value theory. These tools include models such as Markov decision processes, dynamic decision networks, game theory and mechanism design. Bayesian networks are a tool that can be used for reasoning (using the Bayesian inference algorithm), learning (using the expectation-maximisation algorithm), planning (using decision networks) and perception (using dynamic Bayesian networks). Probabilistic algorithms can also be used for filtering, prediction, smoothing, and finding explanations for streams of data, thus helping perception systems analyse processes that occur over time (e.g., hidden Markov models or Kalman filters).

Classifiers and statistical learning methods The simplest AI applications can be divided into two types: classifiers (e.g., "if shiny then diamond"), on one hand, and controllers (e.g., "if diamond then pick up"), on the other hand. Classifiers are functions that use pattern matching to determine the closest match. They can be fine-tuned based on chosen examples using supervised learning. Each pattern (also called an "observation") is labeled with a certain predefined class. All the observations combined with their class labels are

known as a data set. When a new observation is received, that observation is classified based on previous experience. There are many kinds of classifiers in use. The decision tree is the simplest and most widely used symbolic machine learning algorithm. K-nearest neighbour algorithm was the most widely used analogical AI until the mid-1990s, and Kernel methods such as the support vector machine (SVM) displaced k-nearest neighbour in the 1990s. The naive Bayes classifier is reportedly the "most widely used learner" at Google, due in part to its scalability. Neural networks are also used as classifiers.

Artificial neural networks

An artificial neural network is based on a collection of nodes also known as artificial neurons, which loosely model the neurons in a biological brain. It is trained to recognise patterns; once trained, it can recognise those patterns in fresh data. There is an input, at least one hidden layer of nodes and an output. Each node applies a function and once the weight crosses its specified threshold, the data is transmitted to the next layer. A network is typically called a deep neural network if it has at least 2 hidden layers. Learning algorithms for neural networks use local search to choose the weights that will get the right output for each input during training. The most common training technique is the backpropagation algorithm. Neural networks learn to model complex relationships between inputs and outputs and find patterns in data. In theory, a neural network can learn any function. In feedforward neural networks the signal passes in only one direction. The term perceptron typically refers to a single-layer neural network. In contrast, deep learning uses many layers. Recurrent neural networks (RNNs) feed the output signal back into the input, which allows short-term memories of previous input events. Long short-term memory networks (LSTMs) are recurrent neural networks that better preserve longterm dependencies and are less sensitive to the vanishing gradient problem. Convolutional neural networks (CNNs) use layers of kernels to more efficiently process local patterns. This local processing is especially important in image processing, where the early CNN layers typically identify simple local patterns such as edges and curves, with subsequent layers detecting more complex patterns like textures, and eventually whole objects.

Deep learning

Deep learning uses several layers of neurons between the network's inputs and outputs. The multiple layers can progressively extract higher-level features from the raw input. For example, in image processing, lower layers may identify edges, while higher layers may identify the concepts relevant to a human such as digits, letters, or faces. Deep learning has been used to improve the performance of programs in subfields of artificial intelligence including computer vision, speech recognition, natural language processing, image classification and others. Deep neural networks and backpropagation had been in development since 1950. Starting in 2012, the speed of deep learning was increased one hundred-fold by switching to GPUs, and there were enormous amounts of data available on the internet (called at the time "big data") as well as curated datasets used for benchmark testing, such as ImageNet. Usage of deep learning increased in 2012–2015 due to these improvements. In 2025, Geoffrey Hinton said that, until the 2010s, "We couldn't do anything very impressive because we didn't have enough data and we didn't have enough computation."

GPT Generative pre-trained transformers (GPTs) are large language models (LLMs) that generate text based on the semantic relationships between words in sentences. Text-based GPT models are pre-trained on a large corpus of text that can be from the Internet. The pretraining consists of predicting the next token (a token being usually a word, subword, or punctuation). Throughout this pretraining, GPT models accumulate knowledge about the world and can then generate human-like text by repeatedly predicting the next token. Typically, a subsequent training phase makes the model more truthful, useful, and harmless, usually with a technique called reinforcement learning from human feedback (RLHF). Current GPT models are prone to generating falsehoods called "hallucinations". These can be reduced with RLHF and

quality data, but the problem has been getting worse for reasoning systems. Such systems are used in chatbots, which allow people to ask a question or request a task in simple text. Current models and services include ChatGPT, Claude, Gemini, Copilot, and Meta AI. Multimodal GPT models can process different types of data (modalities) such as images, videos, sound, and text.

Hardware and software

In the late 2010s, graphics processing units (GPUs) that were increasingly designed with AI-specific enhancements and used with specialised TensorFlow software had replaced previously used central processing unit (CPUs) as the dominant means for large-scale (commercial and academic) machine learning models' training. Specialised programming languages such as Prolog were used in early AI research, but general-purpose programming languages like Python have become predominant. The transistor density in integrated circuits has been observed to roughly double every 18 months—a trend known as Moore's law, named after the Intel co-founder Gordon Moore, who first identified it. Improvements in GPUs have been even faster, a trend sometimes called Huang's law, named after Nvidia co-founder and CEO Jensen Huang.

Applications

AI and machine learning technology is used in most of the essential applications of the 2020s, including:

search engines (such as Google Search) targeting online advertisements recommendation systems (offered by Netflix, YouTube or Amazon) driving internet traffic targeted advertising (AdSense, Facebook) virtual assistants (such as Siri or Alexa) autonomous vehicles (including drones, ADAS and self-driving cars) automatic language translation (Microsoft Translator, Google Translate) facial recognition (Apple's FaceID or Facebook's DeepFace and Google's FaceNet) image labelling (used by Facebook, Apple's Photos and TikTok).

AI-assisted software development

Chatbots

Physical AI

Health and medicine

AlphaFold 2 (2021) demonstrated the ability to approximate, in hours rather than months, the 3D structure of a protein. In 2023, it was reported that AI-guided drug discovery helped find a class of antibiotics capable of killing two different types of drug-resistant bacteria. In 2024, researchers used machine learning to accelerate the search for Parkinson's disease drug treatments. Their aim was to identify compounds that block the clumping, or aggregation, of alpha-synuclein (the protein that characterises Parkinson's disease). They were able to speed up the initial screening process ten-fold and reduce the cost by a thousand-fold. A 2026 Nature article titled "Dozens of AI disease-prediction models were trained on dubious data" highlighted the use of unreliable data being used to train AI medical prediction models for stroke and diabetes in 125 research articles. Evidence suggested some of the AI tools that were developed on unreliable data had been used on patients, although it was not clear if there were adverse outcomes.

Gaming

Game playing programs have been used since the 1950s to demonstrate and test AI's most advanced techniques. Deep Blue became the first computer chess-playing system to beat a reigning world chess champion, Garry Kasparov, on 11 May 1997. In 2011, in a Jeopardy! quiz show exhibition match, IBM's question answering system, Watson, defeated the two greatest Jeopardy! champions, Brad Rutter and Ken Jennings, by a significant margin. In March 2016,

AlphaGo won 4 out of 5 games of Go in a match with Go champion Lee Sedol, becoming the first computer Go-playing system to beat a professional Go player without handicaps. Then, in 2017, it defeated Ke Jie, who was the best Go player in the world. Other programs handle imperfect-information games, such as the poker-playing program Pluribus. DeepMind developed increasingly generalistic reinforcement learning models, such as with MuZero, which could be trained to play chess, Go, or Atari games. In 2019, DeepMind's AlphaStar achieved grandmaster level in StarCraft II, a particularly challenging real-time strategy game that involves incomplete knowledge of what happens on the map. In 2021, an AI agent competed in a PlayStation Gran Turismo competition, winning against four of the world's best Gran Turismo drivers using deep reinforcement learning. In 2024, Google DeepMind introduced SIMA, a type of AI capable of autonomously playing nine previously unseen open-world video games by observing screen output, as well as executing short, specific tasks in response to natural language instructions.

Mathematics

In January 2025, Microsoft proposed the technique rStar-Math that leverages Monte Carlo tree search and step-by-step reasoning, enabling a relatively small language model like Qwen-7B to solve 53% of the AIME 2024 and 90% of the MATH benchmark problems. Google DeepMind has developed models for solving mathematical problems: AlphaDev, AlphaEvolve, AlphaGeometry, AlphaProof, AlphaTensor, and FunSearch. When natural language is used to describe mathematical problems, converters can transform such prompts into a formal language such as Lean to define mathematical tasks. The experimental model Gemini Deep Think accepts natural language prompts directly and achieved gold medal results in the International Math Olympiad of 2025. In June 2026, an international group of mathematicians published a statement calling on mathematicians, professional organizations, and policymakers to disclose the use of AI, uphold peer review, and protect the openness and values of the discipline. In September 2026, OpenAI announced that one of their models had produced a counter-example to the Navier–Stokes existence and smoothness problem, a Millennium Prize Problem in mathematics. As of 2026, the claim has not been verified by any external mathematicians and is the subject of a priority dispute.

Finance According to Nicolas Firzli, director of the World Pensions & Investments Forum, it may be too early to see the emergence of highly innovative AI-informed financial products and services. He argues that "the deployment of AI tools will simply further automatise things: destroying tens of thousands of jobs in banking, financial planning, and pension advice in the process, but I'm not sure it will unleash a new wave of [e.g., sophisticated] pension innovation."

Military

Military AI usage attempts to enhance command and control, communications, sensors, integration and interoperability. Intelligence collection and analysis, logistics, cyber operations, information operations, and semiautonomous and autonomous vehicles are being researched. AI technologies are used in coordination of sensors and effectors, threat detection and identification, marking of enemy positions, target acquisition, coordination and deconfliction of distributed Joint Fires between networked combat vehicles, both human-operated and autonomous. AI has been used in military operations in Iraq, Syria, Israel and Ukraine.

Generative AI

Agents

AI agents are software entities designed to perceive their environment, make decisions, and take actions autonomously to achieve specific goals. These agents can interact with users, their environment, or other agents. AI agents are used in various applications, including

virtual assistants, chatbots, autonomous vehicles, game-playing systems, and industrial robotics. AI agents operate within the constraints of their programming, available computational resources, and hardware limitations. This means they are restricted to performing tasks within their defined scope and have finite memory and processing capabilities. In real-world applications, AI agents often face time constraints for decision-making and action execution. Many AI agents incorporate learning algorithms, enabling them to improve their performance over time through experience or training. Using machine learning, AI agents can adapt to new situations and optimise their behaviour for their designated tasks.

Web search Microsoft introduced Copilot Search in February 2023 under the name Bing Chat. Copilot Search provides AI-generated summaries. Generative AI features added to Google Search include AI Overviews in May 2024 and AI Mode in May 2025.

Sexuality Applications of AI in this domain include AI-enabled menstruation and fertility trackers that analyse user data to offer predictions, AI-integrated sex toys (e.g., teledildonics), AI-generated sexual education content, and AI agents that simulate sexual and romantic partners (e.g., Replika). AI is also used for the production of non-consensual deepfake pornography, raising significant ethical and legal concerns. AI technologies have also been used to attempt to identify online gender-based violence and online sexual grooming of minors.

Other industry-specific tasks In a 2017 survey, one in five companies reported having incorporated "AI" in some offerings or processes. In the field of evacuation and disaster management, AI has been used to investigate patterns in large-scale and small-scale evacuations using historical data from GPS, videos or social media. During the 2024 Indian elections, US\$50 million was spent on authorised AI-generated content, notably by creating deepfakes of allied (including sometimes deceased) politicians to better engage with voters, and by translating speeches to various local languages. The use of generative AI by law firms for legal research resulted in the creation of the global "AI Hallucination Cases" database, in April 2025, established by HEC Paris and Sciences Po legal data analysis lecturer Damien Charlotin. By 2026, judges had issued sanctions and bar associations had issued warnings due to attorney submissions to the courts containing fabricated case law citations hallucinated by AI tools.

Ethics AI has potential benefits and potential risks. AI may be able to advance science and find solutions for serious problems: Demis Hassabis of DeepMind hopes to "solve intelligence, and then use that to solve everything else". However, as the use of AI has become widespread, several unintended consequences and risks have been identified. In-production systems can sometimes not factor ethics and bias into their AI training processes, especially when the AI algorithms are inherently unexplainable in deep learning.

Risks and harm

Privacy and copyright

Machine learning algorithms require large amounts of data. The techniques used to acquire this data have raised concerns about privacy, surveillance and copyright. AI-powered devices and services, such as virtual assistants and IoT products, continuously collect personal information, raising concerns about intrusive data gathering and unauthorised access by third parties. The loss of privacy is further exacerbated by AI's ability to process and combine vast amounts of data, potentially leading to a surveillance society where individual activities are constantly monitored and analysed without adequate safeguards or transparency. Sensitive user data collected may include online activity records, geolocation data, video, or audio. For example, in order to build speech recognition algorithms, Amazon has recorded millions of private conversations and allowed temporary workers to listen to and transcribe some of them. Opinions about this widespread surveillance range from those

who see it as a necessary evil to those for whom it is clearly unethical and a violation of the right to privacy. AI developers argue that this is the only way to deliver valuable applications and have developed several techniques that attempt to preserve privacy while still obtaining the data, such as data aggregation, de-identification and differential privacy. Since 2016, some privacy experts, such as Cynthia Dwork, have begun to view privacy in terms of fairness. Brian Christian wrote that experts have pivoted "from the question of 'what they know' to the question of 'what they're doing with it'." Generative AI is often trained on unlicensed copyrighted works, including in domains such as images or computer code; the output is then used under the rationale of "fair use". Experts disagree about how well and under what circumstances this rationale will hold up in courts of law; relevant factors may include "the purpose and character of the use of the copyrighted work" and "the effect upon the potential market for the copyrighted work". Website owners can indicate that they do not want their content scraped via a "robots.txt" file. However, some companies will scrape content regardless because the robots.txt file has no real authority. In 2023, leading authors (including John Grisham and Jonathan Franzen) sued AI companies for using their work to train generative AI. Another discussed approach is to envision a separate sui generis system of protection for creations generated by AI to ensure fair attribution and compensation for human authors.

Dominance by tech giants The commercial AI scene is dominated by Big Tech companies such as Alphabet Inc., Amazon, Apple Inc., Meta Platforms, and Microsoft. Some of these companies already own the vast majority of existing cloud infrastructure and computing power from data centres, allowing them to entrench further in the marketplace. For most of 2025-2026, AI-related companies have also dominated stock market returns. According to a July 2024 report from S&P Global, investors had started questioning whether the enormous spending on AI infrastructure and data centers would generate sufficient profits, leading to increased volatility in technology and semiconductor stocks and to concerns about a potential AI bubble. In a 2025 Economic Policy article, economists Anton Korinek and Jai Vipra described some AI related market sectors as highly concentrated.

Power needs and environmental impacts

Technology companies have built electricity and artificial intelligence infrastructure to facilitate the AI boom of the 2020s. A 2025 report from the consulting firm McKinsey & Company estimated that by 2030, \$2.7 trillion would be invested into AI infrastructure and data centres in the US, surpassing World War II's Manhattan Project every month. In January 2024, the International Energy Agency (IEA) released Electricity 2024, Analysis and Forecast to 2026. This is the first IEA report to make projections for data centres and power consumption by AI and cryptocurrency. The report states that power demand for these uses might double by 2026, with the additional power consumption equaling that of Japan. Power consumption by AI is responsible for an increase in fossil fuel use, and has delayed closings of obsolete, carbon-emitting coal energy facilities. A ChatGPT search involves 10 times as much electrical energy as a Google search. A 2024 Goldman Sachs research paper, AI Data Centers and the Coming US Power Demand Surge, found "US power demand (is) likely to experience growth not seen in a generation...." and forecasts that, by 2030, US data centres will consume 8% of US power, as opposed to 3% in 2022, presaging growth for the electrical power generation industry by a variety of means. Data centres' need for more and more electrical power is such that they might max out the electrical grid. The Big Tech companies counter that AI can be used to maximise the utilisation of the grid by all. In 2024, The Wall Street Journal reported that big AI companies have begun negotiations with the US nuclear power providers to provide electricity to the data centres. In March 2024, Amazon purchased a Pennsylvania nuclear-powered data centre for US\$650 million. In September 2024, Microsoft announced an agreement with Constellation Energy to re-open the Three Mile Island nuclear power plant to provide Microsoft with 100% of all electric power produced by the plant for 20 years. Reopening the plant, which suffered a partial nuclear meltdown of its

Unit 2 reactor in 1979, will require Constellation to get through strict regulatory processes which will include extensive safety scrutiny from the US Nuclear Regulatory Commission. If approved (this will be the first ever US re-commissioning of a nuclear plant), over 835 megawatts of power – enough for 800,000 homes – will be produced. The cost for re-opening and upgrading is estimated at US\$1.6 billion and is dependent on tax breaks for nuclear power contained in the 2022 US Inflation Reduction Act. As of 2024, the US government and the state of Michigan have been investing almost US\$2 billion to reopen the Palisades nuclear reactor on Lake Michigan. Closed since 2022, the plant was scheduled to be reopened in October 2025. After the last approval in September 2023, Taiwan suspended the approval of data centres north of Taoyuan with a capacity of more than 5 MW in 2024, due to power supply shortages. Taiwan aims to phase out nuclear power by 2025. Singapore imposed a ban on the opening of data centres in 2019 due to electric power, but lifted it in 2022. Although most nuclear plants in Japan have been shut down after the 2011 Fukushima nuclear accident, according to an October 2024 Bloomberg article in Japanese, cloud gaming services company Ubitus, in which Nvidia has a stake, is looking for land in Japan near a nuclear power plant for a new data centre for generative AI. On 1 November 2024, the Federal Energy Regulatory Commission (FERC) rejected an application submitted by Talen Energy for approval to supply some electricity from the nuclear power station Susquehanna to Amazon's data centre. According to the Commission Chairman Willie L. Phillips, it is a burden on the electricity grid as well as a significant cost-shifting concern to households and other business sectors. In 2025, a report prepared by the IEA estimated the greenhouse gas emissions from the energy consumption of AI at 180 million tons. By 2035, these emissions could rise to 300–500 million tonnes depending on what measures will be taken. This is below 1.5% of the energy sector emissions. The emissions reduction potential of AI was estimated at 5% of the energy sector emissions, but rebound effects (for example if people switch from public transport to autonomous cars) can reduce it. In June 2026, the United Nations University Institute for Water, Environment and Health released a report about cumulative environmental impacts of AI, not only carbon. The report found that AI has significant negative impacts on land, water, and climate, as it uses large amounts of electricity, requires mining of minerals, and generates large amounts of electronic waste. In 2025, data centres consumed more electricity than Saudi Arabia, and ChatGPT alone consumed an amount of water equivalent to the needs of 500,000 people in Sub-Saharan Africa. The negative impacts are not distributed equally across the planet but are concentrated in some places. Data centres are often built in places already suffering from water deficit. For example, AI deployment has already led to an increase in water scarcity in Mexico and Uruguay. It created problems with energy supply in Ireland, which forbade construction of new data centres near Dublin until 2028. Measures can involve tradeoffs, for example, transitioning from coal to bioenergy reduces carbon footprint but increases the impact on land and water. Energy efficiency might not reduce environmental impact, because as AI becomes more efficient, it also becomes cheaper and more widely used.

Misinformation